Pakistan J. Pharmacol. ISSN 0255-7088
Vol. 14(1), January 1997, pp.63-79

REVIEW

REPORT ON THE AVAILABILITY OF SEVEN CONTROLLED PSYCHOACTIVE SUBSTANCES IN PAKISTAN AS RELATED TO REGULATIONS

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1. Terms of Reference

The term of reference of the Consultant were: to conduct a field study in Pakistan to review the effects of regulation on the availability of seven controlled substances and their use. At the discussion at WHO headquarters, it was planned that the two and a half week study in Pakistan, Islamabad and the four provincial capitals would cover the following substances:

a. Under the Single Convention of Narcotic Drugs 1961
Schedule II

Dextropropoxyphene

b. Under the Convention on Psychotropic Substances (1971)
Schedule IV

Phenobarbital
Alprazolam
Diazepam

Schedule III

Pentazocine
Buprenorphine

c. Under the 1988 United Nations Convention Against Ilicit Traffic in Narcotic Drugs and Psychoactive Substances

Ephedrine, pseudoephedrine and their salts

The above proposal was discussed at the Ministry of Health with experts working in the Ministry of Health and in Academic Institutions. The group were of the view that a research project be established with the Ministry of Health, the National Institute of Health, and the provincial health authorities as well as with medical colleges and schools of pharmacy, where students could be encouraged to undertake a project investigating the seven substances in detail (see Annex 1).

It was agreed with Dr. F.R.Y. Fazli, the Drug Controller, that the writer would obtain information from the Ministry of Health (and the provincial capitals) on dextropropoxyphene, ephedrine and pseudoephedrine in detail during this visit. He would undertake a restricted study of the drugs which would be investigated in detail through discussions with the selected individuals i.e. manufacturers, at the pharmacy level, and at the prescribers level. He would also try to identify individuals to undertake the detailed studies as early as possible in coordination with the Ministry of Health. It was hoped that on completion of this study, WHO would be provided with a full report by the Ministry of Health at a later stage.

The writer commenced the mission by having discussions with Dr. Mohamed Ali Barzgar, WHO Representative in Pakistan. In the Ministry of Health, discussions were held with Dr. F.R.Y. Fazli, Drug Controller, when the study was planned, and Dr. Rahan Hafeez of the National Institute of Health who was the co-worker in this study in Rawalpindi and Islamabad. Mr. Shabir Qureshi, Assistant Drug Controller provided the vital statistics on each drug. In Lahore Mr. Khalid Aziz, Federal Drug Inspector, participated in the discussions.

During the site visit to Peshawar, Dr. Mohamed Iqbal, Responsible Office for Medical Stores Depot, accompanied me; Dr. Mehboob Bedini, WHO Operational Officer in Quetta, and Mr. Mohamed Tanweer Alam, Federal Assistant Drug Controller and Mr. Buzdar, Federal Drug Inspector, in Karachi.

2. Control of Dextropropxyhene

Dextropropoxyphene is controlled under Schedule II of the 1961 Convention. In general, substances controlled under Schedule II include drugs more commonly used for medical purposes and needing less rigid control because of the smaller risk of abuse as compared to substances in Schedule I of this Convention for example, morphine.

Preparations of dextropropoxyphene were placed under Schedule III of the 1961 Convention. This includes preparations containing Schedule II drugs, in lower concentrations and in controlled preparations, as well as preparations that have been found by WHO not to be liable to abuse, that cannot produce ill-effects and from which the controlled drug is not recoverable easily. The trade restriction need not apply and the information and statistics required relate only to quantities used in the manufacture of such preparations. The Convention further specified the situation with dextropropoxyphene preparations for oral use containing not more than 135 mg of dextropropoxyphene base per dosage or with a concentration of not more than 2.5% in undivided preparations, provided that such preparations did not contain any substance under the 1971 Convention.

2.1 Situation in Pakistan

In most of the developing countries, it is a common fact that controls after the drug has passed beyond the manufacturer are minimal. Thus the Ministry of Health in Islamabad has decided that dextropropoxyphene 32.5 mg plus 325 mg of paracetamol in a single preparation be allowed for sale. The controls on dextropropoxyphene in Pakistan are:

On receipt of information that the consignment has arrived in the factory, the Ministry of Health requests a production schedule for the manufacturer of the finished drug, along with the distribution record. Thereafter, Federal Drug Inspectors and Assistant Drug Controllers of the regional offices, forward the information to the Provincial Drug Inspectors with the advice to confirm that the drug has been received by the retailers/wholesalers and distributors of the area, and to confirm that they are dispensing the drug according to provincial drug sales rules.

The Pakistan National List of Essential Drugs was first published in May 1994. This list has two narcotics (pethidine and morphine) and five psychotropic drugs (lorazepam, pentazocine, phenobarbital, bupronorphine and diazepam). Dextropropoxyphene has not been included in this list. It had been observed during the last year that certain controlled substances were not available in different parts of the country.

2.2 Availability of dextropropoxyphene

The import of dextropropoxyphene for the last six years in Pakistan was as shown below:

1989	1990	1991	1992	1993	1994
1500 kg	2320 kg	3700 kg	4400 kg	4200 kg	5000 kg

These figures indicate that there has been a steady increase over the last six years. Requests for dextropropoxyphene as received by the Ministry of Health from the manufacturers, reached a level of approximately 25,000 kg recently. Subsequently, the INCB was requested to revise the estimate for Pakistan for this drug from 5100 to 8500 kg in 1995. Whereas in 1993 there were 13 manufacturers of dextropropoxyphene, the number has now increased to 21 during 1995.

3. Control of Ephedrine

These substances have been traditionally used for the preparation of cough syrup in Pakistan for a long time. Ephedrine was manufactured by Marker Alkaloids in Quetta for a number of years. Recently Alphachemical have purchased the plant from Marker alkaloids, and are manufacturing the alkaloids in Lahore.

The 1988 Convention has placed ephedrine and pseudoephedrine and their salts wherever the existence of such salts is possible into Table I. The basis for this was that these substances were used as precursors in the manufacture of stimulants scheduled under the 1971 Convention on Psychotropic Substances. These substances are useful therapeutic agents and their use as precursors in a number of countries was minimal, and control on their export is a further safeguard.

3.1 Situation in Pakistan

Since Pakistan is a signatory to the 1988 Convention, regulations were framed under the present import policy, whereby the import of all substances included in Table I and acetic anhydride (Table II) is allowed strictly in accordance with the requirements of industrial consumers, as determined by the Revenue Division, Government of Pakistan. The Revenue Division determines the requirements submitted by individual pharmaceutical manufacturing units after consultation with the Ministry of Health. The statistics related to this subject are as follows:

July-December 1991	4290 kg
January-June 1992	2825 kg
July-December 1993	2000 kg

After discussions with the concerned pharmaceutical industry and the medical profession, it has been observed that the controls imposed on ephedrine and pseudoephedrine have a negative effect on their supply to meet the needs of patients in Pakistan.

4. Information on the availability of the other five substances

4.1 Phenobarbital

A very useful and cheap drug, it is controlled under Schedule IV of the 1971 Convention.

The number of manufacturers in 1955 was 30; this has risen from 22 in 1992, and 26 in 1993. The manufacturers are allowed to import 100 kg each. It is reported that 20 manufacturers were allowed to import phenobarbital up to July 1995. The remaining manufacturers did not get permission for lack of a consumption report or they obtained import permits after July 1995.

4.2 Diazepam

It is a commonly used tranquilizer which is controlled under Schedule IV of the 1971 Convention.

There are 30 manufacturers for this drug and each is allowed to import 150 kg.

4.3 Pentazocine

It is an analgesic under control of the 1971 Convention, Schedule III.

There are seven manufacturers allowed to import between 5-10 kg. The total amount imported in 1995 was 364 kg; this has risen from 165 kg in 1993, and 340 kg in 1994.

4.4 Buprenorphine

It is an analgesic controlled under Schedule III of the 1971 Convention.

There are seven manufacturers. The maximum amount of imports allowed is 500g per company. Import in 1995 stands at 2,400g; this has risen from 2,300g in 1994 and 115.0g in 1993.

4.5 Alprazolam

It is a tranquilizer controlled under Section IV of the 1971 Convention.

Imports in 1995 were 13kg with five manufacturers and the remaining manufacturers import the drug i its finished form. Figures in 1994 were 14kg and in 1993 4kg with two manufacturers and the remaining were importing the drug in its finished form.

5. Information on the use and abuse of the seven substances

5.1 Dextropropoxyphene

Staff of the Department of Psychiatry Mayo Hospital, Lahore, did not observe a big problem of dependence on this drug. Alao, a large number of psychiatrists in Lahore considered dependence on dextropropoxyphene not enormous. In Quetta, discussions with the Faculty of Medicine at Bolan Medical College, considered that dextropropxyphene has a limited use as an analgesic. In Quetta and Islamabad, oncologists considered dextropropoxyphene as not the main treatment for cancer pain relief. In Rawalpindi, members of the Medical Association who were general medical practitioners, considered dextropropoxyphene as an analgesic and a muscle relaxant. It is useful for pain and stiffness originating from the relaxant. It is useful for pain and stiffness originating from the muscular skeletal system, for example osteo arthritis.

They did not consider it important enough for use in terminal cancer pain, nor did they consider its use for coughs and cold advisable, because it can cause dependence, though its abuse is not common. It is adequately available for use in the following six preparations containing dextropropoxyphene:

Distalgesic	32.5 mg
Wigesic	65 mg
Darvin	32.5 mg
Deprogesic	65 mg
Diagesic	32.5 mg
Algaphan	25 mg

All are combined with paracetamol. These preparations are not available in the Army Medical Services.

A survey of the promotional material from a number of manufacturers indicates the following:

These preparations are advertised as analgesics, antipyretics and spasmolytics. Indications for use are: traumatic pain, headache, toothache spasmodic pain. It is not recommended for use concerning anesthesia and respiratory distress because it depresses the respiratory centre. It should not be used together with alcohol. It has addiction as adverse effects especially when the doses are increased.

A WHO publication entitled "Cancer Pain Relief, 1986" describes dextropropoxyphene as being effective in pain management, particularly in patients with advanced cancer diseases. Its use in one of the priorities in a comprehensive WHO Cancer Programme. Drug therapy is the mainstay of cancer pain management. The basic list contains dextropropoxyphene as one of the main drugs.

5.2 Ephedrine and Pseudophedine

An anesthetist in Lahore informed the writer that ephedrine injectable preparations are useful for spinal anesthesia, but it was not available. However, the tablets are available and so ar the cough syrups containing these drugs.

5.3 Buprenorphine and pentazocine

These drugs are available in certain pharmacies but a number of them require prescriptions before dispensing, especially in Lahore. The psychiatrists in Lahore consider that pentazocine is used by surgeons and oncologists. Unfortunately, general practitioners also use these for patients with pain who become dependent on these drugs. They have seen buprenorphine dependence as prevalent. On average, four new cases of pentazocine and buprenorphine dependence are admitted to Mayo Hospital each month.

A pharmacy in Karachi informed the writer that addicts seek buprenorphine injections from his pharmacy from time to time.

A psychiatrists in Karachi has seen buprenorphine and pentazocine being abused. Members of the Pakistan Medical Association in Rawalpindi consider buprenorphine and pentazocine as commonly used drugs in pre- and post-operative surgical procedures. They are also used for cardiac pain and pain of a colicky nature. They both cause vomiting while pethidine does not. Abuse of buprenorphine and pentazocine is common among the local inhabitants, including doctors and nurses. These drugs are commonly used in terminal cancer pain management.

5.4 Alprazolam

The Rawalpindi General Hospital has treated 135 dependent on this drug. It was registered two years ago and is prescribed widely for anxiety and depression. It is more expensive than diazepam which may limit its abuse. A psychiatrist in Karachi has seen two to three new cases of alpazolam dependence per month in his limited practice. The manufacturer targets a tired housewife and an executive. There is over-use by general practitioners and specialists other than psychiatrists like cardiologists and neurologists. Xenox is one of the trade names and it is referred to as "butterfly pills". The industry's promotion is very high and they give up to 30% rebate to pharmacies instead of the usual 15%. They also promote trips to exotic lands for doctors. Medical Association members in Lahore consider alprazolam freely available. Psychiatrists in Lahore consider this as the most abused drug today. In a gynaecological practice in Lahore, every fourth woman has used this drug.

5.5 Diazepam

It is a good muscle relaxant but its use and dependence have both decline these days. A psychiatrist describes diazepam abuse as out of fashion now. Heroin use is still combined with diazepam. In Lahore, diazepam is freely available, even as an injection. Some pharmacies give a few tablets of diazepam as a gift it you buy, for example, oxytetracycline.

5.6 Phenobarbital

In Rawalpindi, phenobarbital tablets are available but not in injectable form. It is a cheap drug which is ideal for treating epilepsy, eclampsia and fits in children. Certain types of resistant coughs also improve with its use. Its abuse today is referred to as habituation. Phenobarbital abuse is rare in Karachi. It costs 60 rupees per 1,000 tablets with retailers, but wholesalers demand 185 rupees instead.

6. Recommendations

6.1 Availability of "MSD continuous"

It is good news for oncologists that a new analgesic "MSD continuous" containing 30mg of morphine manufactured by ASTA, Germany, is available for use. This costs 60 rupees per tablet which is still an exorbitant price. The oncologists now have relief to treat terminal cancer pain, since it is understood that Pakistan cannot utilize the raw opium grown around its frontiers for such an essential drug because of international restrictions on it. The availability of this drug must be expedited.

6.2 Availability of Controlled Drugs only at specific pharmacies

The Pakistan Medical Association in Lahore and Rawalpindi consider strengthening prescription control on drugs as a very important step. It is the key to solving the drug misuse and abuse problem in Pakistan. It should only be selected pharmacies who may sell narcotic and psychotropic drugs with immediate effect.

6.3 Strengthening of Undergraduate Medical Education

Undergraduate medical education needs to be strengthened. Pakistan Medical and Dental Council (PMDC) must play an active role in this urgent task. Re-registration of doctors in Pakistan every five years is another area which requires active collaboration between PMDC and the Health Services Academy at Islamabad.

6.4 Views of the Pharma Buraux

The Pharma Bureaux are an association of multinational companies which operate in Pakistan. The Bureaux consolidate requests by the manufacturers to receive quotas of controlled substances - but a different approach is needed. The substances and their quantities must depend on the report of an Inspection Team, consisting of officials assigned to drug control and experts from academic. Future needs of a manufacturer should be established by past performances.

The resources of the Federal Government today in the form of Federal Drug Inspectors is only eight for the whole country. These facilities must be improved otherwise the whole exercise would be simply theoretical.

The Association is willing to offer a single fellowship to an internist with post-graduate qualifications (e.g. FCPS) to be sent for one-year's training in clinical pharmacology, for example, in the University of Aberdeen. They will offer this Fellowship to the Medical School in Larkana or Peshawar.

The Association of Medical Directors can be a useful faculty to collaborate with PMDC in organizing training for doctors in the rational use of drugs during the re-

registration activities. The industry has promised to take the first step by approaching the PMDC.

6.5 Pakistan Pharmaceutical Manufacturing Association's views

The Pakistan Pharmaceutical Manufacturing Association and its President have observed that the placing of ephedrine and pseudo- ephedrine under the 1988 UN Convention, which resulted in the application of control measures similar to those for acetic anhydride by the Government of Pakistan, as regrettable. As a consequence, the Revenue Division become involved in their control, which has been seen as a step in the wrong direction. Ephedrine is a useful therapeutic agent and its export can be rigidly controlled to safeguard other countries. In Pakistan, diversion of ephedrine into the illicit manufacture of methamphetamien seems impossible. It is advised that in future, controls on ephedrine and pseudo- ephedrine be made similar to other psychoactive drugs, which is primarily the responsibility of the Ministry of Health. The writer strongly supports this plea.

6.6 Recommendations regarding the use of dextropropoxyphene

The writer appreciates the position taken by the Ministry of Health to limit the quantity of dextropropoxyphene in combination products to 32.5 mg, which has resulted in keeping the incidence of dependence on dextropropoxyphene to a low level so far. The promotional material by the manufacturers must be reviewed and should exclude unjustified and sometimes dangerous material. The use of dextropropoxyphene for the treatment of patients dependent on opiates must be avoided. Headache is a dangerous indication for use. Dextropropoxyphene is a dependence-producing substance and its combination with paracetamol has reduced its dependence liability, but has improved its therapeutic effects as an analgesic. This fact must be brought to the attention of the doctors and the patients. Contra-indications and side effects must be clearly stated, for example, combination with alcohol and psychotropic drugs must be avoided. Injectable forms of preparations must be stopped. The Single Convention forbids such preparations.

It is recommended that clinical researchers in Pakistan debate the merit of this drug in future scientific meetings, basing the debate on their own experiences.

6.7 Annex 3 relates to a proposal originating from the leaders of the medical profession in Quetta, the Director-General of Health, and the officer commanding CMH, to undertake a study of prescription monitoring on the lines of experience gained in Rawalpindi and Islamabad. A similar proposal has also been made to the National Institute of Health, and the Ministry of Health at Islamabad by Peshawar were four hospitals, that is the Hayat Sherpo Hospital, Lady Reading Hospital, CMH and Police Hospital will participate in this type of activity.

7. Summary and Conclusions

Pakistan is signatory to the three existing international drug control treaties. Pakistan has a population of 130 million people and has a turnover of 20 billion rupees of pharmaceuticals yearly. The resourcesavailable for drug control has been repeatedly considered to be inadequate by the medical profession, the pharmaceutical profession and the pharmaceutical industry. A serious effort on the part of the Government will be required to overcome this deficiency. Pakistan shares with other developing countries an urgent need for the tightening up of pharmaceutical monitoring available to the society through Government efforts and academia.

Certain activities must be promoted, for example, clinical pharmacological expertise, adverse drug reaction monitoring, drug utilization studies, study of prescription patterns by doctors, and strengthening facilities for management of cases of acute intoxification by drugs and other environmental pollutants. Pakistan has identified these as priority areas. The Ministry of Health, National Institute of Health and the Pakistan College of Physicians and Surgeons have realized that by working together they can produce results early in these areas.

Another important area is to strengthen undergraduate medical education and to inform doctors about new developments in drug therapy by getting the Pakistan Medical and Dental Council and Health Services Academy involved in these educational processes. Pharmacy education also needs to be geared to serve the interests of the patients and the doctors through the modification of their syllabus as advised by WHO.

In 1990, the writer and Dr. Fazli investigated the availability of diazepam and phenobarbital. It was found that 32 manufacturers were each receiving 200 kg of phenobarbital, and 23 manufacturers each receiving 200 kg of diazepam yearly. It was the observation then that these two drugs were imported in excess of Pakistan's needs. It was also observed that certain quantities of both the substances went into the illegal market and were combined with heroin to meet the needs of the local addicts, and some were smuggled abroad. The present study has shown an improvement because the combined use of heroin with the two substances has declined significantly. Today, there are 30 manufacturers each of whom are allowed to import 100 kg of phenobarbital, while a similar number of manufacturers import 150 kg each of diazepam.

Pakistan has effectively controlled morphine and pethidine. They are only available as injections through the government hospitals. General practitioners do find it inconvenient at times to go through the control process, and thus select other alternatives which are easily available in the private pharmacies. Liquid morphine for oral use can also be procured through the Ministry of Health and Excise Department at

Lahore. The urgent need for morphine tablets for terminally ill cancer patients has now been met by registering a product imported from Germany.

Since the 1971 Convention came into force in 1976, countries are continuing to struggle and place under effective control psychotropic drugs, and in particular the benzodiazepines bupronorphine and pentazocine. Pakistan is not an exception to this phenomenon. The present regulations state that when a manufacturing unit receives its quota, the Ministry of Health requests a production schedule for the manufacture of the finished drug and a distribution record. Thereafter, the federal drug inspectors, and ADCs of the Regional Offices forward this information to the Provincial Drug Inspectors with the advice to confirm that the drug has been received by the retailer/wholesaler and distributors of the area, and also to confirm that they are dispensing the drug according to the Provincial Drug Rules.

Availability of drugs without prescription is of great concern to the medical and pharmaceutical professions. Today licenses to sell are issued even to those who do not have a degree in pharmacy. Thus there is a lack of professional responsibility of the personnel responsible for pharmacies in the private sector, which is a significant hindrance towards the rational use of drugs. There is also a discussion going on at the moment to allow the sale of drugs only to selected pharmacies in the private sector who already have qualified staff, and could undertake responsibility for complying with regulations, especially prescription control.

Pakistan is riddled with drug abuse with about 2% of the population abusing opiates, cannabis and synthetic psychoactive drugs. The reduction in demand for illicit drugs is the responsibility of the Narcotics Control Ministry. The educational efforts, essential for producing awareness among the health care professionals, is an important subject to be promoted. Similarly, to equip health care professional for the management of drug dependence is also a priority area. The Government must realize that although large numbers of addictions in Pakistan are through opiates, i.e. opium and heroin, synthetic drugs such as tranquilizers, have contributed towards the morbidity and mortality that we are concerned with. Despite extensive resources available with the Narcotics Control Ministry, we have yet to see cooperation between this Ministry and the Ministry of Health in creating awareness among health care professionals to fight this great menace.

Ephedrine and pseudoephedrine are both useful drugs which the medical profession in Pakistan has been using for a long time as cough remedies. The recent restrictions on these drugs of being controlled more rigidly than psychotropic drugs as possible precursors of methamphetamine, has had a negative effect on the capability of the medical profession. The quantity of imported ephedrine was reduced by nearly half (to 2000 kg) in 1993. The manufacturers' request for a reduction of the present controls to be on par with other psychotropic drugs seems reasonable. Their opinion is that it should be the Ministry of Health's responsibility for allocating quotas and monitoring the distribution of use, and thus excluding the role of the revenue division. This should be accompanied by rigid controls on the exportation of these products abroad.

Of the other psychotropic drugs, alprazolam has proved to be a drug of abuse and this requires the attention of the Ministry of Health in the form of a tightening up of its quota, to reduce promotional activities of the industry and to inform patients and doctors of the dangers associated with its misuse and abuse.

The key to the problem of misuse and abuse of narcotic and psychotropic drugs in Pakistan is in the hands of the Ministry of Health. They have to take a decision concerning the restriction of the availability of controlled drugs to selected pharmacies who only have qualified staff, and need to make these drugs available only on prescription of doctors. This is a unanimous request of the medical profession, pharmacists and the public. Based on personal observations during the three visits to Pakistan in the last six months, the writer also considers that the time for action has come to implement this recommendation.

Annex 1

PROPOSAL TO INVESTIGATE THE AVAILABILITY AND USE OF THE SEVEN SUBSTANCES IN A DETAILED STUDY

A study has been proposed whereby the seven substances can be investigated in greater detail regarding their availability, use and abuse. The study will be carried out in Rawalpindi, Islamabad and Peshawar. This study will be at the divisional level where at least 100 pharmacies will be randomly selected for survey. The study will be carried out by the provincial drug inspectors with support from senior students of pharmacy and/or medicine as part of a community based programme.

The objectives of this study would be:

A. To verify at the *manufacturers level* records of manufacture, quality control and sale of this drug. This would be under the mandate of the Provincial Drug Inspectors in collaboration with the Federal Inspector of Drugs.

B. To investigate at the *pharmacy level* availability, sale records and prescription records. This could be carried out by the provincial drug Inspectors with the assistance of the students of pharmacy and/or medicine as part of their community medicine programme.

C. To ascertain at the *prescribers level* ethical criteria used in writing such prescriptions. The preferred modus shall be by conducting drug utilization studies involving at least 30 major practices. This could also include a survey of 10 detoxification centres to ascertain the proportion of patients addicted to the substances being investigated.

D. To carry out a survey of *traditional medicines* being dispensed in outlets and to investigate the presence of selected psychotropic substances in these products.

The results can be made available to WHO through Dr. F.R.Y. Fazli, Drug Controller, as and when available.

Annex 2

INDICATIONS FOR USE OF DEXTROPROPOXYPHENE PREPARATIONS

The following indications have been listed in various promotional material for dextropropoxyphene preparations. It has been assigned as having analgesic, antipyretic and antispasmodic properties. Its mode of action is at the CNS, skeletal muscles and by inhibiting prostaglandins synthesis.

Indications

Biliary colic	Headache
Renal colic	Toothache
Postoperative pains	Dysmenoshea
	Migraine
	Muscle ache
	Joint pains
	Backache

Additional advantages listed are to afford quick onset with longer duration of action. Efficacy well marked. Non-addictive. Relieves anxiety and tension. Adjunct therapy for controlling narcotic withdrawal symptoms.

One of the preparations is Algaphan which contains 25 mg dextropropoxyphene, and 300mg paracetamol, 1-3 tablets advised for adults in a single dose. To control severe pain 1-2 tablets every 4-6 hours are advised.

Contra indications; Hypersensitivity to dextropropoxyphene and paracetamol. Side effects - mild nausea in isolated cases when doses of more than 6 tablets per day are taken. Not to be given to children. Not to be combined with alcohol and psychotropic drugs. This reduces reaction time and thus is dangerous while driving and for those who need concentration.

Algaphan is available in tablets, ampoules and suppository form.

Annex 3

PROPOSAL FOR A DRUG UTILIZATION STUDY IN SANDEMAN PROVINCIAL HOSPITAL AND COMBINED MILITARY HOSPITAL, QUETTA

Following a visit by the writer to the Directorate of Health at Quetta on 22 January 1996, the following proposal was formulated.

Subsequent to a successful study of this type in the Islamabad/Rawalpindi area, it was proposed that similar study be conducted in Quetta. This would include the analysis of one hundred prescriptions covering both in-patients and outpatients in the following specialities in each of the two hospitals:

Internal medicine, general surgery, paediatrics and psychiatry. The in-patient study will be retrospective while the outpatient study will be prospective.

Dr. Abdul Haq Lasi, Director General Health Services, Baluchistan, was very positive and would encourage such a study. The objectives of the study would be to review the prescribing practices of doctors and fill the gaps for improvement in the future. This exercise could be extended to other specialities within these two hospitals as well as to other hospitals in Baluchistan.

In discussions with Brigadier M. Akram, Commanding Officer, C.M.H., Quetta, he considered the project useful for improving prescribing practices of doctors but prior to agreeing to participation in this project, explicit permission would have to be sought from the Medical Directorate of the Army Medical Corps, Rawalpindi.

WHO has an explicit interest in this type of activity and would be able to offer technical and advisory support to this project on request.

The following persons have agreed to participate in this study:

1. Professor Abdul Malik as Coordinator
2. Dr. Mehboob Ali, Assistant Professor of Medicine
3. Dr. Abdul Bari, Assistant Professor Paediatrics
4. Dr. Habibullah, Assistant Professor Pharmacology
5. Dr. Abdul Rashid, Associate Professor Surgery

Brigadier M. Akram will nominate his colleagues. The writer would keep in touch with developments of the project from Geneva, Switzerland. Dr. Mahboob Badini, WHO Operations Officer, Baluchistan, would be available to the group to obtain technical and advisory support from WHO, and other day-to-day activities. After obtaining the formal approval from the Government of Baluchistan, a communication needs to be made with Dr. F.R.Y. Fazli, Drugs Controller, Ministry of Health at Islamabad about this project. Dr. Syed Mohsin Ali, Executive Director, N.I.H., can provide technical support to this project on request by seconding technical staff to advise this group.

A group from Peshawar have also initiated a similar study in four hospitals, that is the Hayat Sherpo University Hospital, Lady Reading Hospital, Combined Military Hospital and Police Hospital.

Annex 4

LIST OF INSTITUTIONS VISITED AND PERSONS MET

1. Peshawar

Professor Mohamad Ashfaq, Dean, Khyber Medical College
Professor Ziaul Islam Director, Hayat Sherpo University Teaching Hospital
Members of the Faculty of Medicine and Pharmacy
Mr. Muhamed Yunus Khan, Health Secretary of NWFP
Dr. Azmet Afridi, Director, General Health, NWFP
Dr. Mohamed Iqbal, Chief, Medical Store Depot

2. Quetta

Professor A. Malik, Department of Psychiatry and other members of the Faculty of Medicine, Bolan Medical College
Professor M. Ilyas, Director, Centre for Nuclear Medicine and Radiotherapy and Colleagues
Dr. A. Haq Lasi, Director, General Health Services, Baluchistan and his col-

leagues in drug control.
Brigadier M. Akram, Commanding Officer, CMH

3. Lahore

Dr. Yasmin Raashid, Pakistan Medical Association
(Secretary-General) and colleagues
Professor Rashid Chaudry and other Psychiatrists in Lahore
Professor of Psychiatry and Paediatrics, K.E. Medical College
Professors Ijaz Tarein and Tariq Bhutta
Mr. Amjad Ali Jawa, Chairman, Pakistan Pharmaceutical Manufacturers Association and Dr. A.Q. Khokhar, Managing Director, Pakistan, Remington Pharmaceutical Industries Ltd.

4. Rawalpindi

Professor Malik Mubashar and Dr. Khalid Said of the WHO Collaborating Centre in Mental Health, Rawalpindi Medical College
Pakistan Medical Association, Rawalpindi
Dr. Sajjad Minhas, President, and Colleagues

5. Islamabad

Mr. Matiullah, Joint Secretary and Mr. Ilyas Khattak, Narcotics Control Ministry
Dr. N.A. Kazilbash, Institute of Nuclear Medicine, Oncology and Radiotherapy

6. Karachi

Mr. Zafar Moraj, Pharma Bureau
Mr. Shafi Uddin Feroz, Elfroze Chemical Industry and his colleagues Tajmedicos Pharmacy
Mr. Talaat Habib Malik

PNS Shifa, Surgeon Commander, Mowadat Rna
Jinnah Post Graduate Centre Medical Centre
Professor I.H. Bhati, Director
Professor Musarat Hussein, Psychiatry ad
Dr. Surayya Jabeen Cheeaa, Deputy Director (Drugs)
Pakistan College of Physicians and Surgeons
Professor Sultan Farooqui and Professor Akhlaz Un Nabi Khan Agha Khan University
Professor John Kirk, Dean, Faculty of Health and Mebers of the Academic Council

Annex 5

ACKNOWLEDGEMENTS

The writer wishes to express his gratitude to the following persons for their support in the planning and implementation of this consultanteeship: Dr. F.R.Y. Fazli, Drug Controller, Ministry of Health, Government of Pakistan.

Dr. M.A. Barzgar, the WHO Representative in Pakistan

Dr. Rehan Hafiz, Senior Scientific Officer at the National Institute of Health in Islamabad.

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